

Course Instructor: **Nishat Tasnim**  
Course: **CSE103 (Structured Programming)**

Lab: **01**

## Topics Today

1. Learn standard code writing with comments.
2. Variable declaration and assignment.
3. Basic input/output.
4. Basic input/output of different datatype.
5. Basic arithmetic operations.

## Problem 1

Assign your name, age and student id in three separate variables and print them.

## Problem 2

Write a C program to print the following characters in reverse.

| Sample Input  | Sample Output             |
|---------------|---------------------------|
| 'E', 'S', 'C' | The reverse of ESC is CSE |

## Problem 3

Write a C program to print a block F using the hash (#), where the F has a height of six characters and a width of five and four characters.

| Sample Output                      |
|------------------------------------|
| #####<br>#<br>#<br>#####<br>#<br># |

## Problem 4

Write a C program to input two numbers and perform all arithmetic operations (sum, difference, product, quotient and modulus of two given numbers).

| Sample Input                         | Sample Output                                                             |
|--------------------------------------|---------------------------------------------------------------------------|
| First number: 10<br>Second number: 5 | Sum = 15<br>Difference = 5<br>Product = 50<br>Quotient = 2<br>Modulus = 0 |

## Problem 5

Write a C program to input length and width of a rectangle and find area of the given rectangle.

| Sample Input                       | Sample Output                    |
|------------------------------------|----------------------------------|
| Enter length: 5<br>Enter width: 10 | Area of rectangle = 50 sq. units |

## Problem 6

Write a C program to input the radius of a circle from the user and calculate its diameter, circumference, and area.

| Sample Input     | Sample Output                                                              |
|------------------|----------------------------------------------------------------------------|
| Enter radius: 10 | Diameter = 20 units<br>Circumference = 62.79 units<br>Area = 314 sq. units |